

1. Write a program to store 15 integer values in an ArrayList and remove all duplicate elements without using another collection.

```
import java.util.ArrayList;
```

```
public class Main {
```

```
    public static void main(String[] args) {
```

```
        ArrayList<Integer> list = new ArrayList<>();
```

```
        list.add(10);
```

```
        list.add(20);
```

```
        list.add(30);
```

```
        list.add(10);
```

```
        list.add(40);
```

```
        list.add(20);
```

```
        list.add(50);
```

```
        list.add(60);
```

```
        list.add(30);
```

```
        list.add(70);
```

```
        list.add(80);
```

```
        list.add(90);
```

```
        list.add(100);
```

```
        list.add(80);
```

```
        list.add(50);
```

```
        for (int i = 0; i < list.size(); i++) {
```

```
            for (int j = i + 1; j < list.size(); j++) {
```

```
                if (list.get(i).equals(list.get(j))) {
```

```
                    list.remove(j);
```

```

        j--;
    }
}
}

System.out.println(list);
}
}

```

Output

```

Main.java
1- import java.util.ArrayList;
2-
3- public class Main {
4-     public static void main(String[] args) {
5-
6-         ArrayList<Integer> list = new ArrayList<>();
7-
8-         list.add(10);
9-         list.add(20);
10-        list.add(30);
11-        list.add(10);
12-        list.add(40);
13-        list.add(20);
14-        list.add(50);
15-        list.add(60);
16-        list.add(30);
17-        list.add(70);
18-        list.add(80);
19-        list.add(90);
20-        list.add(100);
21-        list.add(80);
22-        list.add(50);
23-
24-        for (int i = 0; i < list.size(); i++) {
25-            for (int j = i + 1; j < list.size(); j++) {
26-                if (list.get(i).equals(list.get(j))) {
27-                    list.remove(j);
28-                    j--;
29-                }
30-            }
31-        }
32-
33-        System.out.println(list);
34-    }
35-}

```

input

[10, 20, 30, 40, 50, 60, 70, 80, 90, 100]

...Program finished with exit code 0
Press ENTER to exit console.

2. Store employee IDs in a LinkedList and display them in both forward and reverse order.

```

import java.util.LinkedList;

public class Main {

    public static void main(String[] args) {

        LinkedList<Integer> list = new LinkedList<>();
    }
}

```

```

list.add(101);

list.add(102);

list.add(103);

list.add(104);


System.out.println("Forward:");

for (int i : list)

    System.out.println(i);


System.out.println("Reverse:");

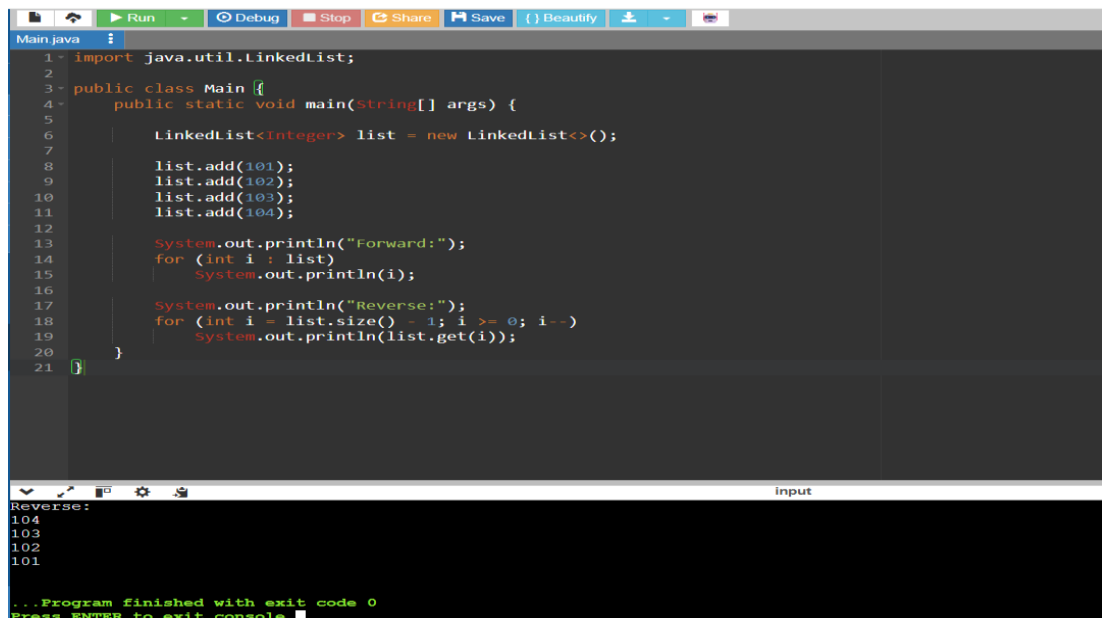
for (int i = list.size() - 1; i >= 0; i--)

    System.out.println(list.get(i));

}

}

```



The screenshot shows an IDE window with a Java file named 'Main.java'. The code in the file is as follows:

```

1 import java.util.LinkedList;
2
3 public class Main {
4     public static void main(String[] args) {
5
6         LinkedList<Integer> list = new LinkedList<>();
7
8         list.add(101);
9         list.add(102);
10        list.add(103);
11        list.add(104);
12
13        System.out.println("Forward:");
14        for (int i : list)
15            System.out.println(i);
16
17        System.out.println("Reverse:");
18        for (int i = list.size() - 1; i >= 0; i--)
19            System.out.println(list.get(i));
20    }
21 }

```

Below the code editor, the console output is visible. It shows the results of the program execution:

```

Reverse:
104
103
102
101
...Program finished with exit code 0
Press ENTER to exit console.

```

3. Store email IDs entered by the user in a HashSet. Display only unique email IDs and count the total unique entries.

```
import java.util.*;

public class Main {

    public static void main(String[] args) {

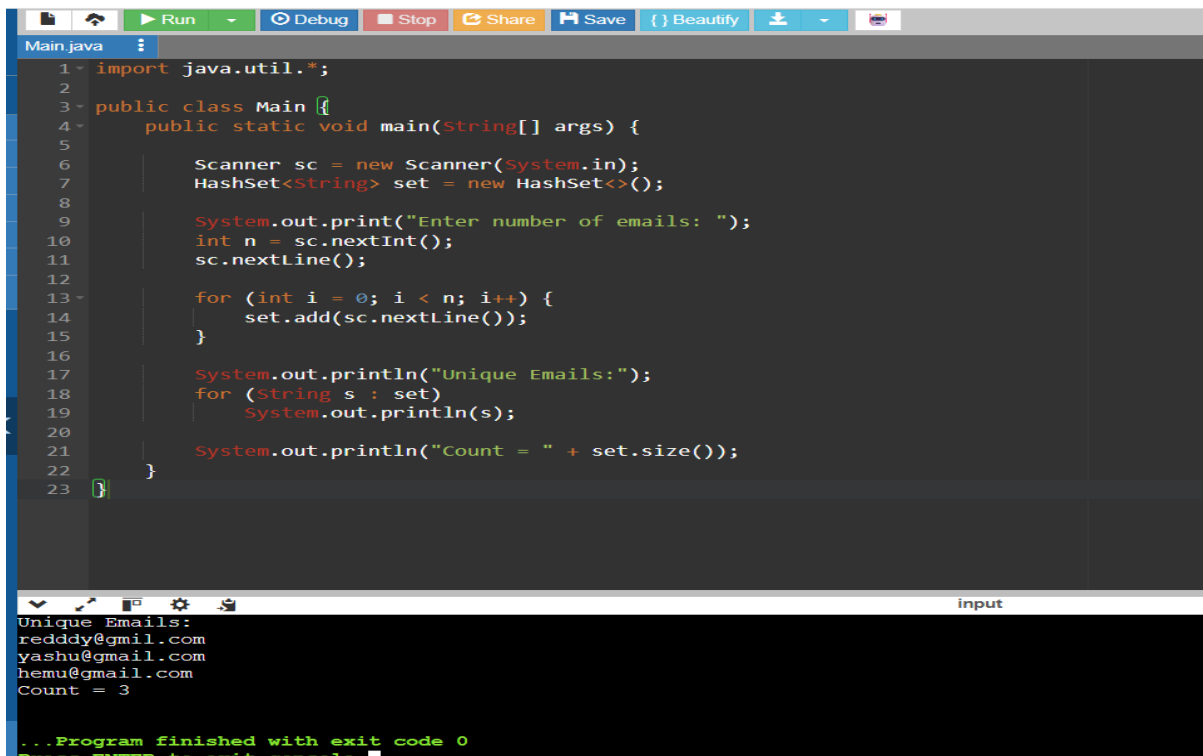
        Scanner sc = new Scanner(System.in);
        HashSet<String> set = new HashSet<>();

        System.out.print("Enter number of emails: ");
        int n = sc.nextInt();
        sc.nextLine();

        for (int i = 0; i < n; i++) {
            set.add(sc.nextLine());
        }

        System.out.println("Unique Emails:");
        for (String s : set)
            System.out.println(s);

        System.out.println("Count = " + set.size());
    }
}
```



```
1 import java.util.*;
2
3 public class Main {
4     public static void main(String[] args) {
5
6         Scanner sc = new Scanner(System.in);
7         HashSet<String> set = new HashSet<>();
8
9         System.out.print("Enter number of emails: ");
10        int n = sc.nextInt();
11        sc.nextLine();
12
13        for (int i = 0; i < n; i++) {
14            set.add(sc.nextLine());
15        }
16
17        System.out.println("Unique Emails:");
18        for (String s : set)
19            System.out.println(s);
20
21        System.out.println("Count = " + set.size());
22    }
23 }
```

Unique Emails:
redddy@gmail.com
yashu@gmail.com
hemu@gmail.com
Count = 3

...Program finished with exit code 0
Press ENTER to exit console

4. Write a Java program to count the frequency of each word in a paragraph using a HashMap.

```
import java.util.*;

public class Main {
    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        System.out.println("Enter paragraph:");
        String str = sc.nextLine();

        String[] words = str.split(" ");

        HashMap<String, Integer> map = new HashMap<>();

        for (String w : words) {
            if (map.containsKey(w))
                map.put(w, map.get(w) + 1);
            else
                map.put(w, 1);
        }
    }
}
```

```

    for (String key : map.keySet()) {
        System.out.println(key + " = " + map.get(key));
    }
}
}
}

```

The screenshot shows an IDE window titled 'Main.java' with the following Java code:

```

1 import java.util.*;
2
3 public class Main {
4     public static void main(String[] args) {
5
6         Scanner sc = new Scanner(System.in);
7
8         System.out.println("Enter paragraph:");
9         String str = sc.nextLine();
10
11         String[] words = str.split(" ");
12
13         HashMap<String, Integer> map = new HashMap<>();
14
15         for (String w : words) {
16             if (map.containsKey(w))
17                 map.put(w, map.get(w) + 1);
18             else
19                 map.put(w, 1);
20         }
21
22         for (String key : map.keySet()) {
23             System.out.println(key + " = " + map.get(key));
24         }
25     }
26 }

```

Below the code editor, the 'input' tab shows the execution output:

```

^?java is easy
= 1
is = 1
easy = 1
java = 1

...Program finished with exit code 0
Press ENTER to exit console

```